

# HEMSHANKAR LAUGI

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## SUMMARY

Honors Biomedical Engineering senior with 2+ years of interdisciplinary research experience spanning machine learning, neuroimaging, computer vision, and biomedical data analysis. Experienced in developing interpretable machine learning models for Alzheimer's disease, analyzing structural MRI data, and applying 3D computer vision to depth imaging and point-cloud analysis. Interested in developing machine learning and computer vision methods for medical imaging and clinically relevant biomedical applications.

## EDUCATION

**B.S. (Honors) in Biomedical Engineering | University of Texas at Arlington (UTA)**

Graduation: May 2027

Minors: Mathematics, Physics

**GPA: 3.96/4.00**

**Honors Thesis:** *Interpretable Machine Learning of Plasma Proteomics Reveals Stage-Specific Signatures Across the Alzheimer's Disease Continuum*

**Thesis Advisor:** Dr. Xi Zhu

**Relevant Coursework:** Calculus I-III, Linear Algebra & Differential Equations, Molecular Biology, Chemistry I-II, Physics I-II, Statistical Inference, Human Physiology, Linear Systems, Bioinformatics, Biomolecular Engineering, Medical Imaging, Bioinstrumentation, AI in Medicine, Digital Signal Processing

## EXPERIENCE

**Undergraduate Research Assistant | Biomedical Research in AI & Neuroimaging (BRAIN) Lab, UTA**

Oct 2025 – Present

- Led a plasma proteomics study from literature-driven identification of research gaps through computational analysis and biological interpretation.
- Designed a machine learning pipeline for multi-stage Alzheimer's classification using 146 plasma analytes from 566 ADNI participants.
- Evaluated 5 models with nested stratified 5x5 cross-validation and covariate adjustment, achieving stage-specific AUCs up to 0.963.
- Utilized SHAP interpretability to identify stage-specific proteomic signatures linked to immune, metabolic, inflammatory, and vascular pathways.
- Applied multivariable regression models to investigate protein-brain relationships by mapping plasma biomarkers to structural MRI measures.

**Undergraduate Researcher | NSF Research Training Group (RTG), Department of Mathematics, UTA**

Sep 2025 – Present

- Co-developed an ordinary differential equation (ODE) model to simulate multidrug antibiotic resistance dynamics in *S. aureus* and *P. aeruginosa*.
- Estimated model parameters from experimental growth and antibiotic susceptibility data using nonlinear optimization in MATLAB.
- Synthesized literature to guide modeling of resistance through spontaneous mutation, back mutation, horizontal gene transfer, and fitness costs.
- Calibrated models to evaluate effects of multiple resistance mechanisms and treatment schedules on bacterial clearance and resistant populations.

**Process Development Intern | Amgen, Thousand Oaks, CA**

May 2026 – Aug 2026

- Developed an automated cell culture scale-up workflow with 10X increase in volume throughput and reduction of up to 50% in manual resources.
- Optimized multi-plate sampling and passaging methods in an automated liquid handler, enabling 3X faster processing.
- Validated process accuracy, consistency, and no cross-contamination during automated passaging via automated cell counter and flow cytometry.
- Presented findings in weekly meetings, documented observations in electronic lab notebook (ELN), and prepared a job aid for workflow execution.

**Analytical Science Intern | Boehringer Ingelheim, Fremont, CA**

May 2025 – Aug 2025

- Developed a highly reproducible MauriceFlex icIEF method that offers a 3X reduction in fractionation method development compared to CEX-HPLC.
- Achieved 45% reduction in process run time by deploying Supersonic icIEF method, streamlining protein characterization workflows.
- Improved vendor's peak prediction algorithm through empirical data analysis and collected charge fractions using 25% fewer reagents and time.
- Operated 17+ experiments to establish fractionation method, optimizing 10+ parameters to collect fractions with high purity and concentration.

**Undergraduate Research Assistant | Cancer Immunotherapy Lab, UTA**

May 2024 – Apr 2025

- Conducted molecular, genetic, and protein studies involving nanobody development for immune checkpoint blockade, biochemical assays to understand interaction between TCR mimic antibodies and peptide MHC, and yeast surface display model for rapid antibody characterization.
- Contributed to a machine learning algorithm that predicts the binding specificity of nanobodies to cancer-specific peptide complex.
- Documented experimental workflows and results to ensure data integrity, reproducibility, and efficient troubleshooting of experimental failures.

## PROJECTS

### **Handheld Facial Asymmetry Assessment Device** | *Software, UI/UX & Computer Vision Lead*

Aug 2026 – Present

- Developing a 3D computer vision pipeline using OAK-D depth imaging and point clouds for objective assessment of facial asymmetry.
- Implementing 3D facial landmark localization and XYZ coordinate extraction to quantify spatial differences in facial features.
- Evaluating depth accuracy, measurement repeatability, and motion artifacts using multi-frame acquisition and controlled-object validation.

### **Densely Connected Neural Network with Average Pooling Transformation** | *Machine Learning Researcher*

Aug 2025 – Feb 2026

- Designed DenseNext, a dense neural-network architecture integrating average-pooling and residual mappings to improve parameter efficiency.
- Collaborated with a Ph.D. researcher on architectural design, parameter optimization, model validation, and manuscript preparation.
- Achieved accuracy comparable to DenseNet on CIFAR-10, CIFAR-100, and SVHN datasets with ~30% fewer parameters.

## PRESENTATIONS

- **Laugi, H.**, et al. “*From Plasma-Based Classification to Structural Neuroimaging Correlates Across the Alzheimer’s Disease Continuum.*” UT Systemwide Brain Research Summit 2026, Austin, TX, Nov 2026. **Accepted for Poster Presentation.**
- **Laugi, H.**, et al. “*Evaluating Multidrug Treatment Strategies for Polymicrobial Coinfection in Cystic Fibrosis.*” Tenth International Conference on Mathematical Modeling & Analysis of Populations in Biological Systems (ICMA-X), University of Maryland, College Park, MD, Oct 2026. **Accepted for Oral Presentation (30 min).**
- **Laugi, H.**, Hosseinzadehketilath, M., & Zhu, X. “*Interpretable Machine Learning of Plasma Proteomic Biomarkers for Multi-Stage Alzheimer’s Disease Classification.*” Honors Research Symposium, University of Texas at Arlington, Arlington, TX, Apr 2026.
- **Laugi, H.**, & Mehra, R. “*Establishing Imaged Capillary Isoelectric Focusing (icIEF) Fractionation on MauriceFlex for High-Throughput Analytical Characterization of Protein Charge Variants.*” Summer Internship Presentation, Boehringer Ingelheim, Fremont, CA, Aug 2025.
- **Laugi, H.**, Ghaffari, S., & Weidanz, J. “*Engineering a Yeast Surface Display Platform for Peptide/MHC Expression and Rapid Characterization of T Cell Receptor Mimic (TCRm) Antibodies.*” College of Engineering Innovation Day, University of Texas at Arlington, Arlington, TX, Apr 2025.
- **Laugi, H.**, Ghaffari, S., & Weidanz, J. “*Assessing the Binding Specificity of Nanobodies to Major Histocompatibility Complex (MHC)-E Protein.*” THIA Conference, Arlington, TX, Sep 2024.

## SKILLS

- **Programming:** Python, MATLAB, R, Linux/Unix
- **Quantitative & Computational:** Statistical analysis, machine learning, mathematical modeling, ordinary differential equations (ODEs), model interpretability, nonlinear optimization, multivariable regression
- **Biomedical Imaging & Data:** Clinical, multi-omics, and neuroimaging data analysis, computer vision, 3D point clouds, depth imaging
- **Tools & Software:** LabVIEW, BIOPAC, DSI Studio, SolidWorks, SoftMax Pro, Empower (Waters), UNIFI (Waters)
- **Experimental Methods:** Cell culture (suspension & adherent), flow cytometry, capillary isoelectric focusing (icIEF), western blotting, capillary electrophoresis, PCR, BCA assay, affinity chromatography, protein-binding assays, transfection, yeast surface display

## AWARDS

- **Alfred R. and Janet H. Potvin Outstanding Bioengineering Student Award** | *UTA Bioengineering Department*, Apr 2025
- **Third Place – Poster Presentation** | *UTA College of Engineering Innovation Day*, Apr 2025
- **Second Place – Research Presentation** | *TExBioMed 2024 Competition*; Awarded full sponsorship to attend *THIA 2024 Conference*, CIHI | Aug 2024

## LEADERSHIP & ADDITIONAL EXPERIENCE

### **Co-Founder & Vice-Chair** | *IEEE Engineering in Medicine and Biology Society (EMBS), UTA Chapter*

Jan 2026 – Present

- Co-founded UTA’s first IEEE EMBS student chapter, securing organizational approval and building a 9-member interdisciplinary executive board.
- Lead technical and professional-development initiatives, including bioinformatics workshops and biomedical industry/lab tours for 20+ students.
- Developing a faculty-mentored research initiative to form student teams, match them with faculty based on interests, and facilitate projects.

### **Research Volunteer** | *Cook Children’s Neuroscience Research Center, Fort Worth, TX*

Aug 2025 – Dec 2025

- Analyzed cerebral palsy data via DSI Studio tractography, focusing on diffusion MRI, fiber tract reconstruction, and structural connectivity analysis.
- Participated in patients’ data collection sessions to understand various neuroimaging modalities and clinical data acquisition workflows.